Digital Forensics Laboratory

EXPERIMENT NO: 01

TITLE: Comprehensive Digital Evidence Acquisition and Command-Line Analysis using FTK Imager, WinPmem, and Volatility

1. AIM

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager, capturing live volatile memory (RAM) using the WinPmem command-line utility, and analyzing the extracted memory dump using Volatility 3 commands via the PassMark Volatility Workbench.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tools:**
    - AccessData FTK Imager (GUI for non-volatile physical drive imaging)
    - WinPmem (winpmem_mini_x64_rc2.exe) (CLI tool for volatile memory capture)
    - Volatility 3 Framework (vol.exe / PassMark Volatility Workbench V3.0)
- **Hardware:** Evidence PC/Workstation, Target Storage Drive (hp v221w 16GB USB Device).

3. THEORY

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.

- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).
- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., -f <image_path> windows.pslist.PsList) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

4. PROCEDURE

Part A: Acquiring Non-Volatile Memory (Disk Image) using FTK Imager

Step 1: Select the Source Type

Open FTK Imager as an Administrator. Navigate to **File** > **Create Disk Image**. In the Select Source dialog, choose **Physical Drive**.

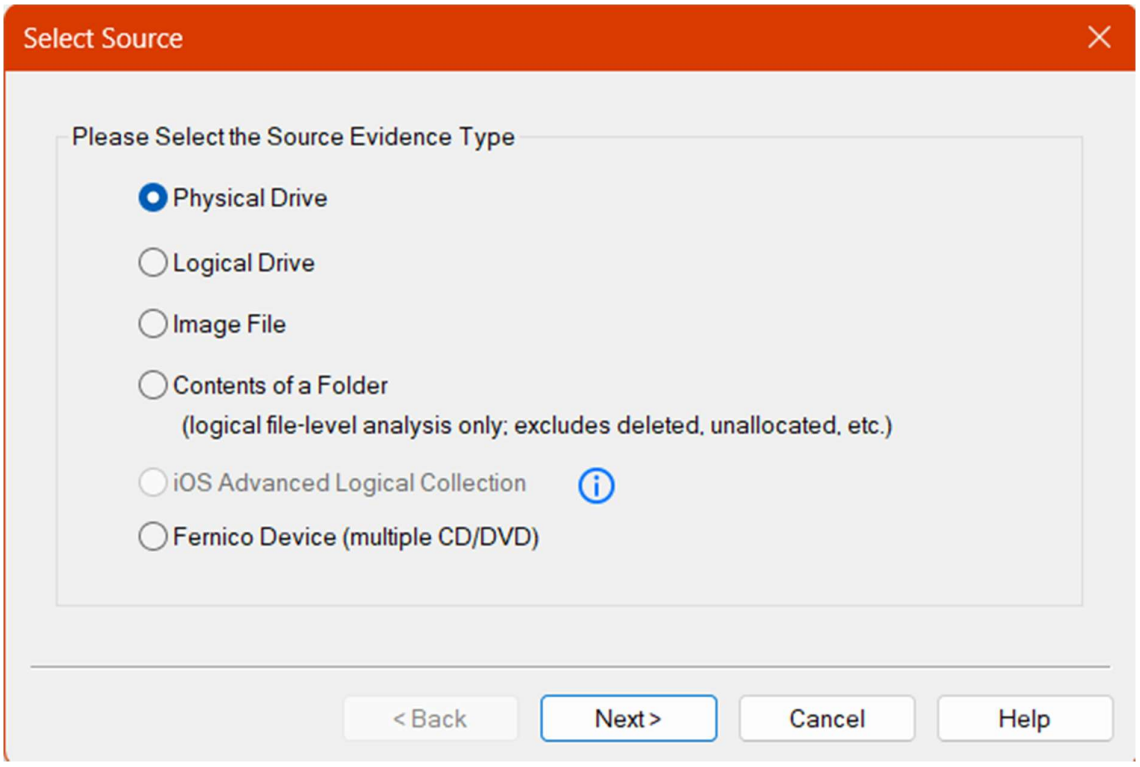


Figure 1: Selecting Physical Drive as the source evidence type

Step 2: Select the Specific Drive

Select the target evidence drive (\\.\PHYSICALDRIVE1 - SanDisk Cruzer Blade USB Device [30GB USB]) from the dropdown.

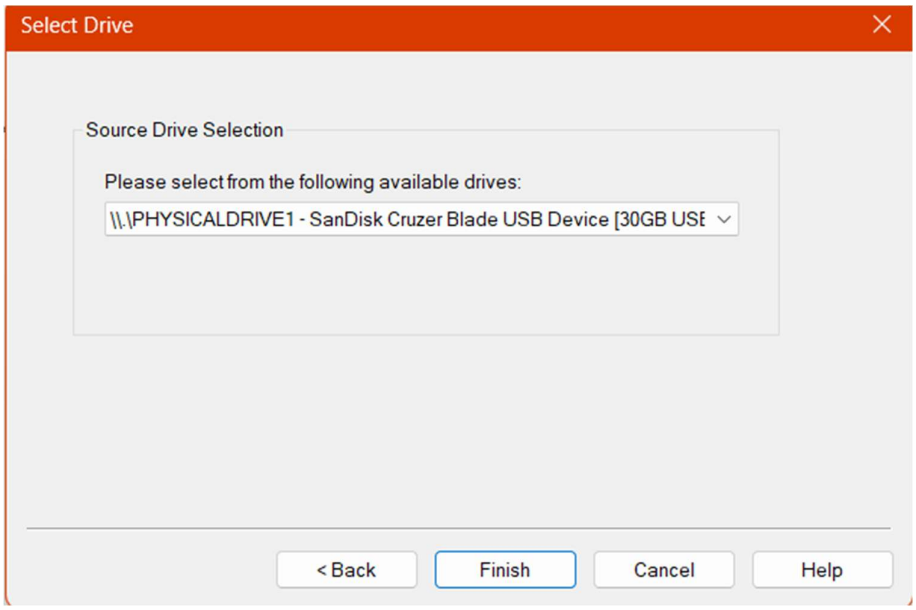


Figure 2: Selecting the target USB device for acquisition

Step 3: Choose the Image Type

Select **Raw (dd)** as the destination image type to create a pure bitstream copy.

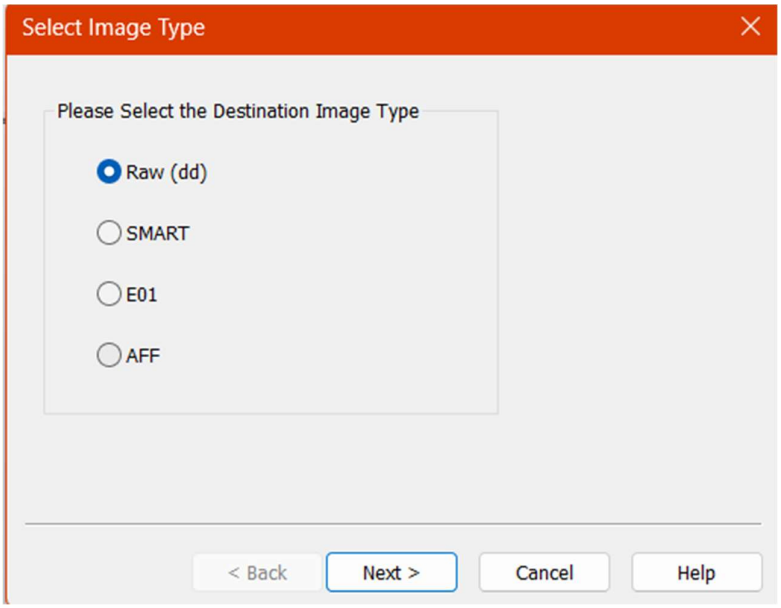


Figure 3: Configuring the destination image format to Raw (dd)

Step 4: Enter Evidence Item Information

Provide the standardized case details to maintain the chain of custody:

- **Case Number:** 99240040143
- **Evidence Number:** 1
- **Unique Description:** Examine
- **Examiner:** Ragul Nagaraj

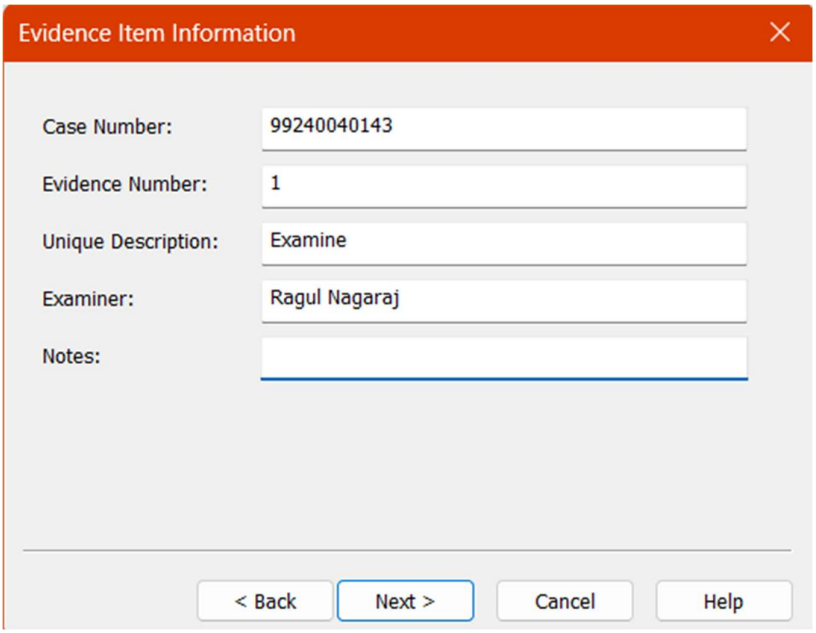


Figure 4: Entering the forensic case and examiner details

Step 5: Configure Image Destination

Set the destination folder and the filename to Result. Set the Fragment Size to 20 MB and check **Verify images after they are created**.

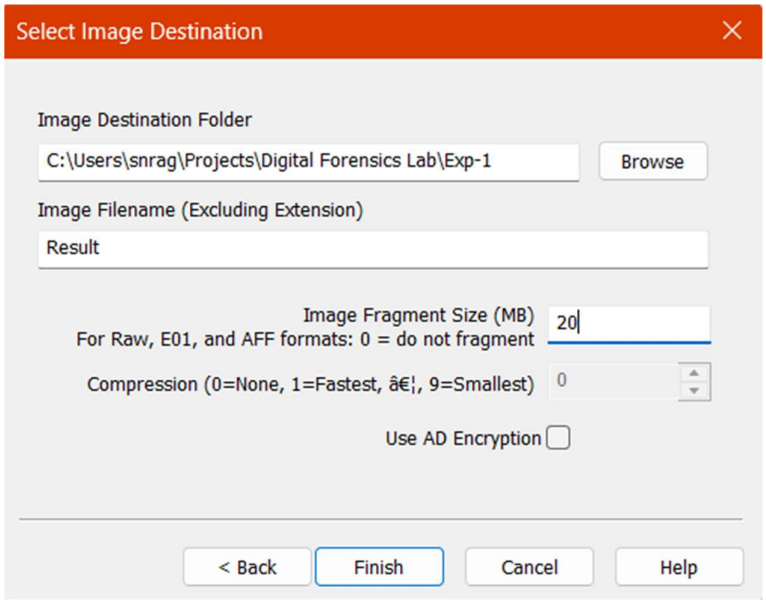


Figure 5: Setting the destination path, filename, and fragmentation size

Step 6: Monitor Acquisition Progress & Hash Verification

Click **Start**. Once complete, FTK Imager hashes the original drive and the newly created image. Review the Drive/Image Verify Results.

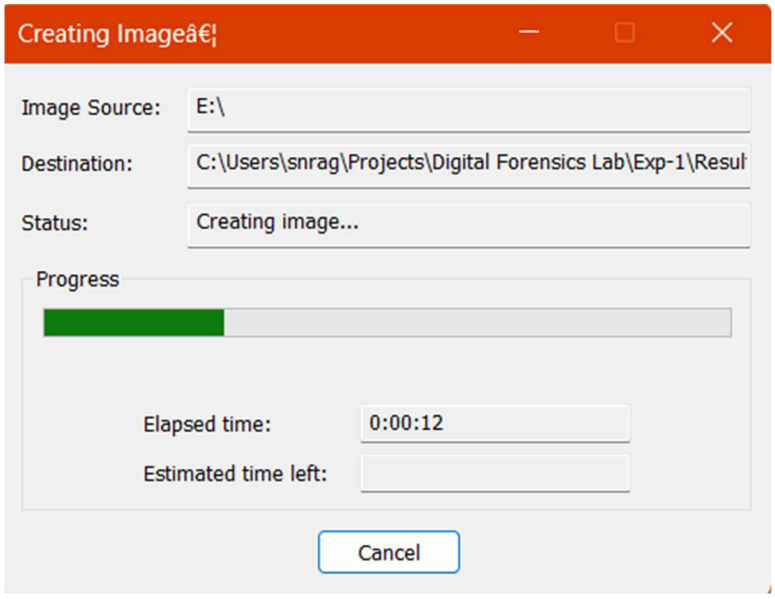


Figure 6: Disk imaging progress tracking

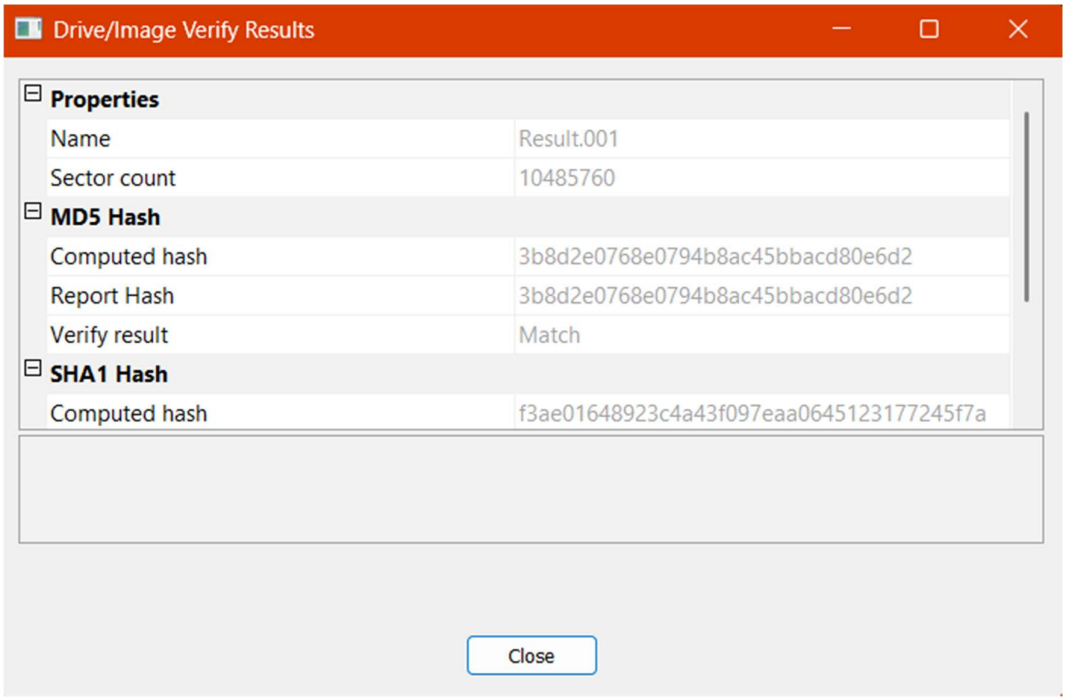


Figure 7: Successful verification of MD5 and SHA1 hashes

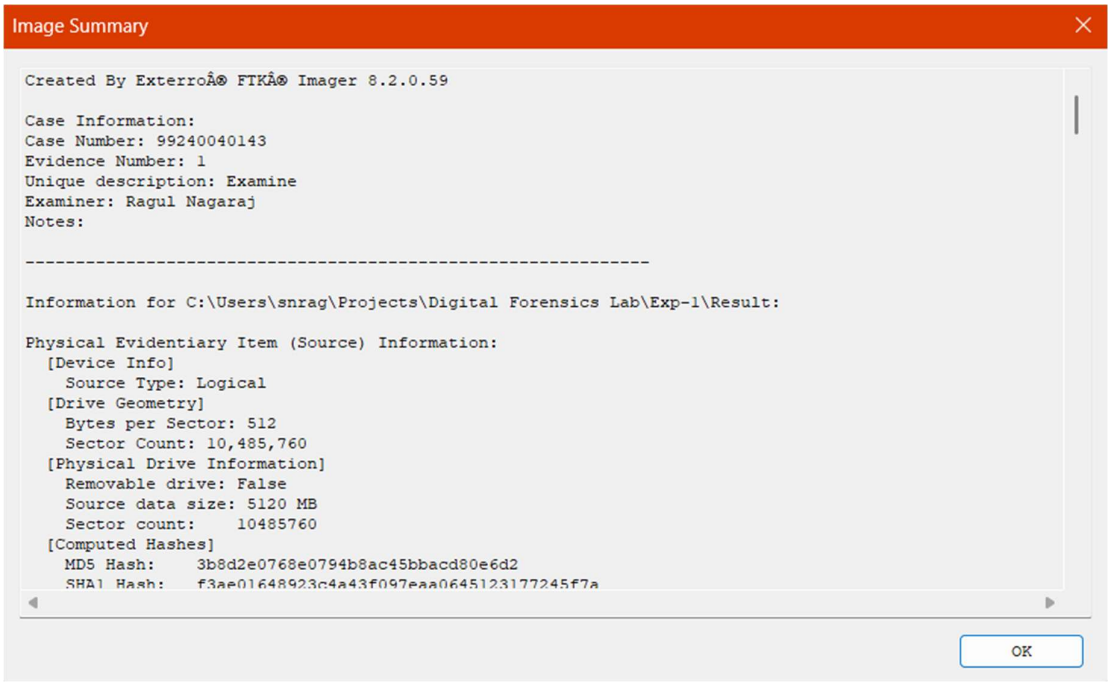


Figure 8: Image Summary detailing the physical device geometry

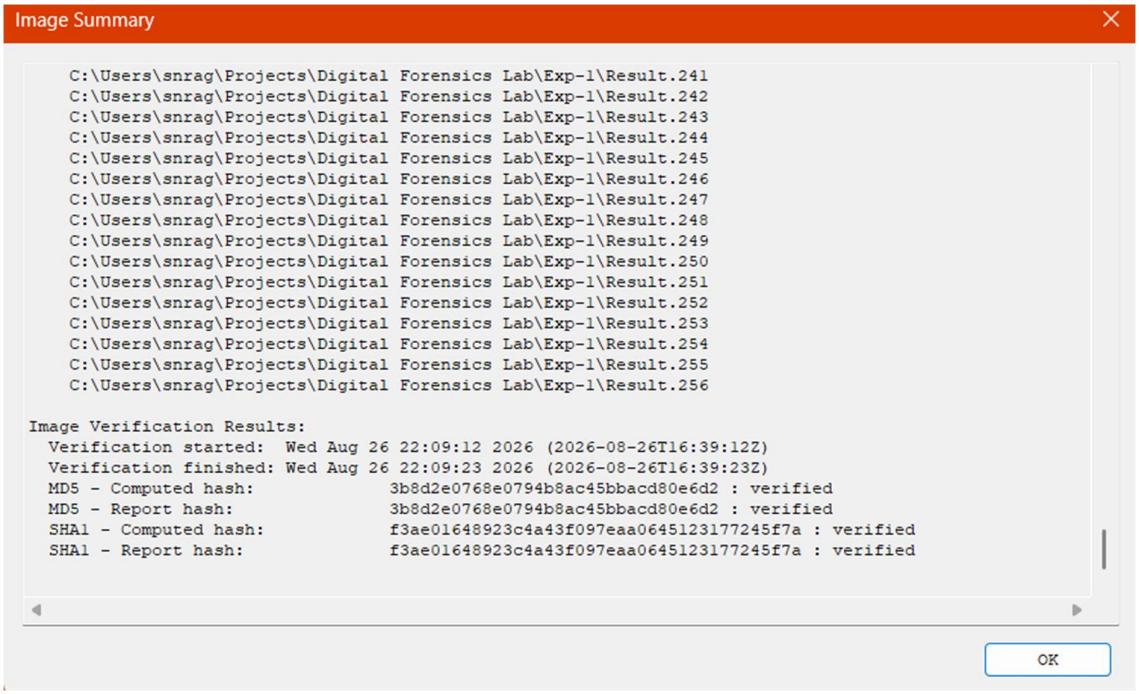


Figure 9: Image Summary detailing the hash verification results and segment list

Part B: Acquiring Volatile Memory (RAM) using WinPmem CMD

Step 1: Launch Command Prompt

Press Win + R, type cmd, and press Ctrl + Shift + Enter to open the Command Prompt as an **Administrator**.

Step 2: Navigate to Directory

Use the cd command to navigate to the folder containing the WinPmem executable

Step 3: Check Available Parameters

Execute the help command to view the WinPmem usage parameters:

DOS

winpmem_mini_x64_rc2.exe -h

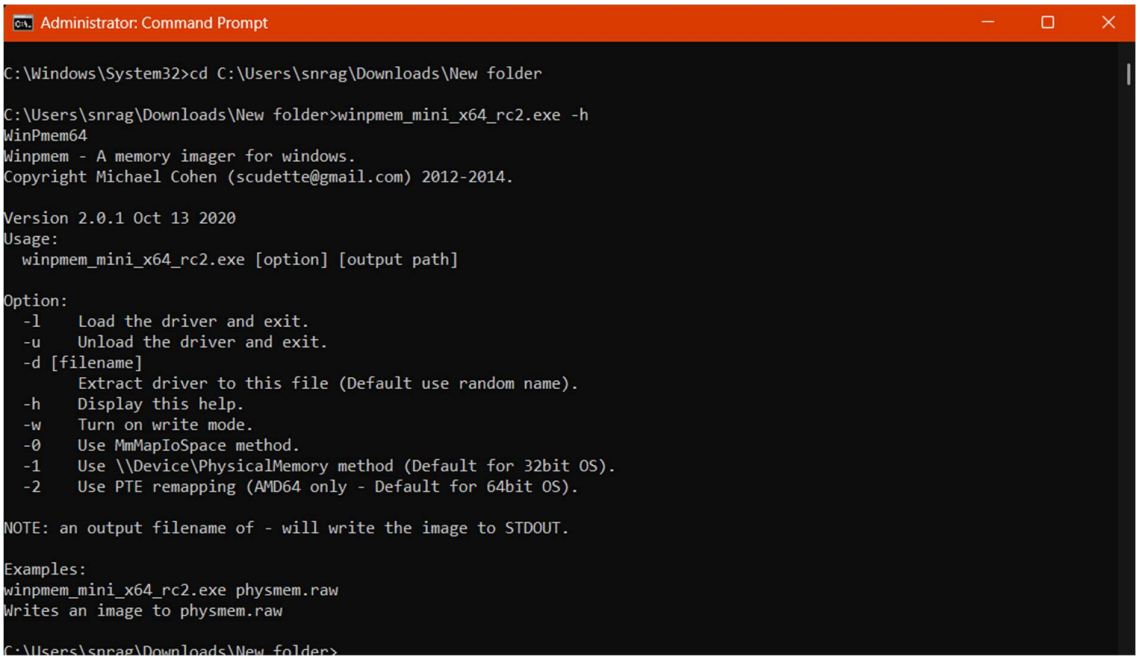


Figure 10: Viewing the WinPmem help menu and available parameters

Step 4: Execute Memory Capture Command

Run the memory extraction command, specifying the output filename ragul.raw. The utility will map the physical memory and write it to the file:

DOS

winpmem_mini_x64_rc2.exe ragul.raw

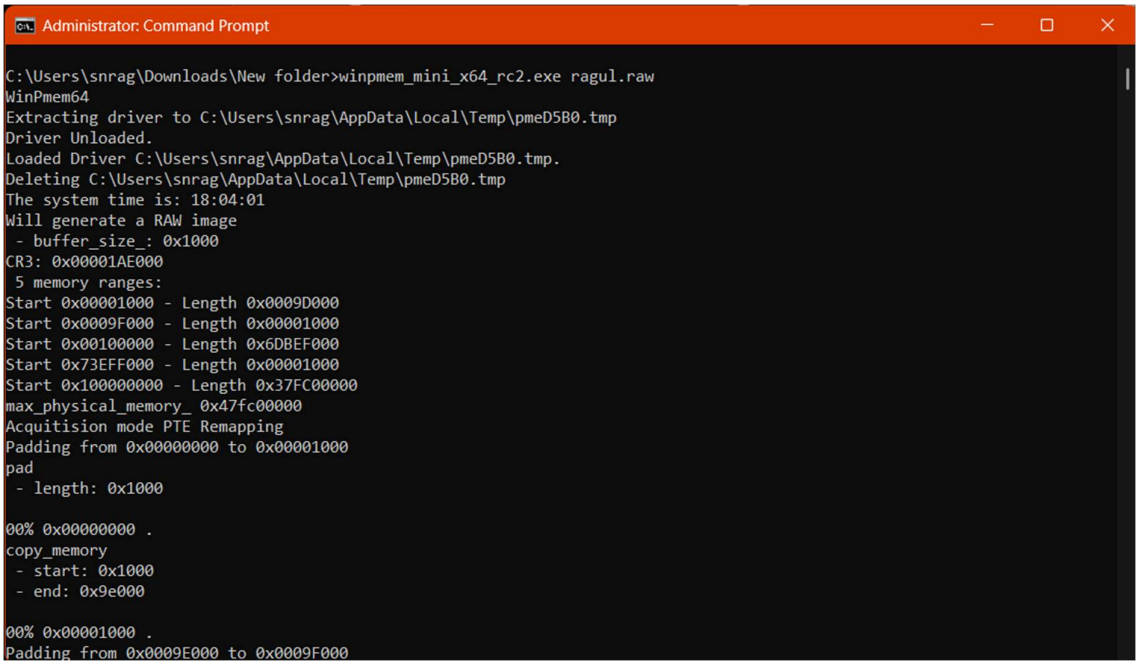


Figure 11: Initializing WinPmem and loading the extraction driver

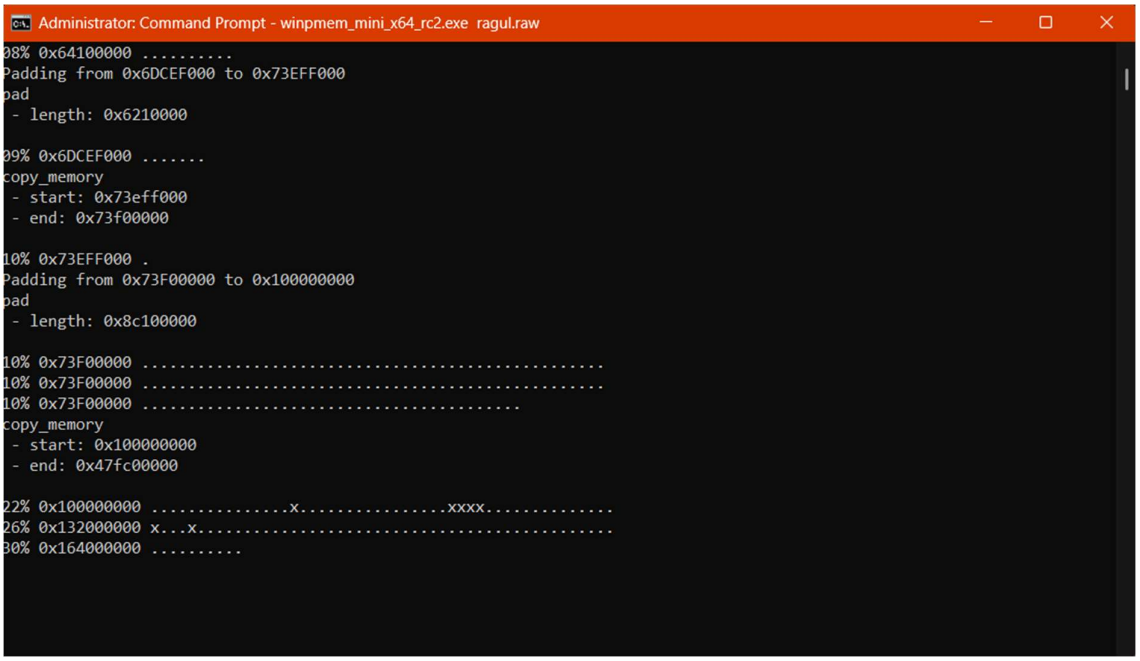


Figure 12: Memory capture in progress

Step 5: Verification of Dump

Wait for the terminal to display Driver Unloaded. Navigate to the directory to verify the 18.2 GB ragul.raw file.

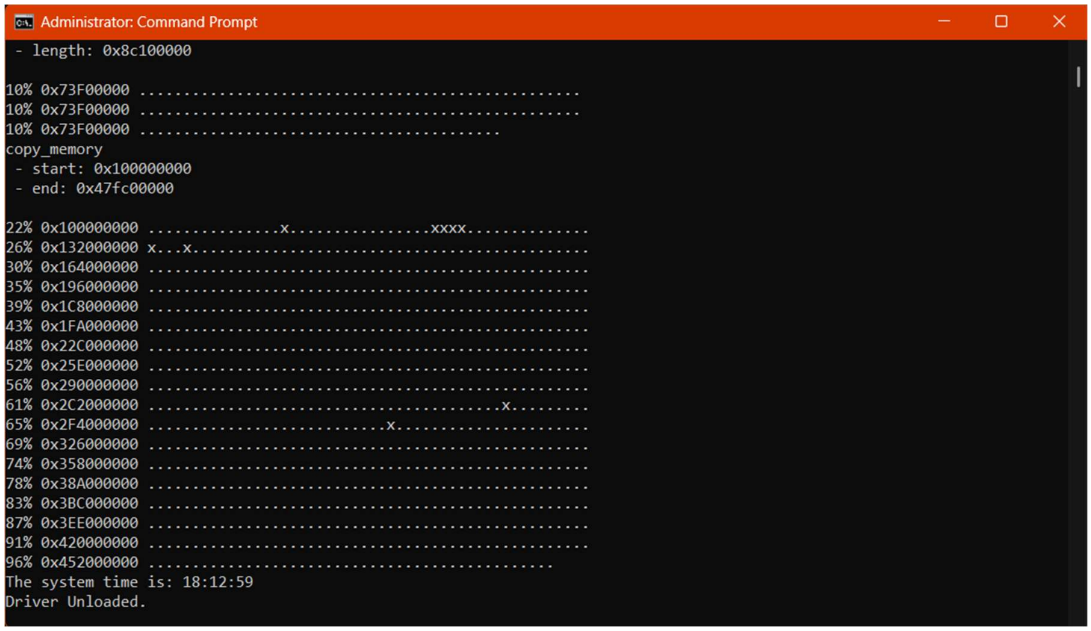


Figure 13: Successful completion of the volatile memory capture

Part C: Volatile Memory Analysis using Volatility CLI / Workbench

Step 1: Launch Volatility Environment

Open PassMark Volatility Workbench (which acts as a GUI frontend for vol.exe).

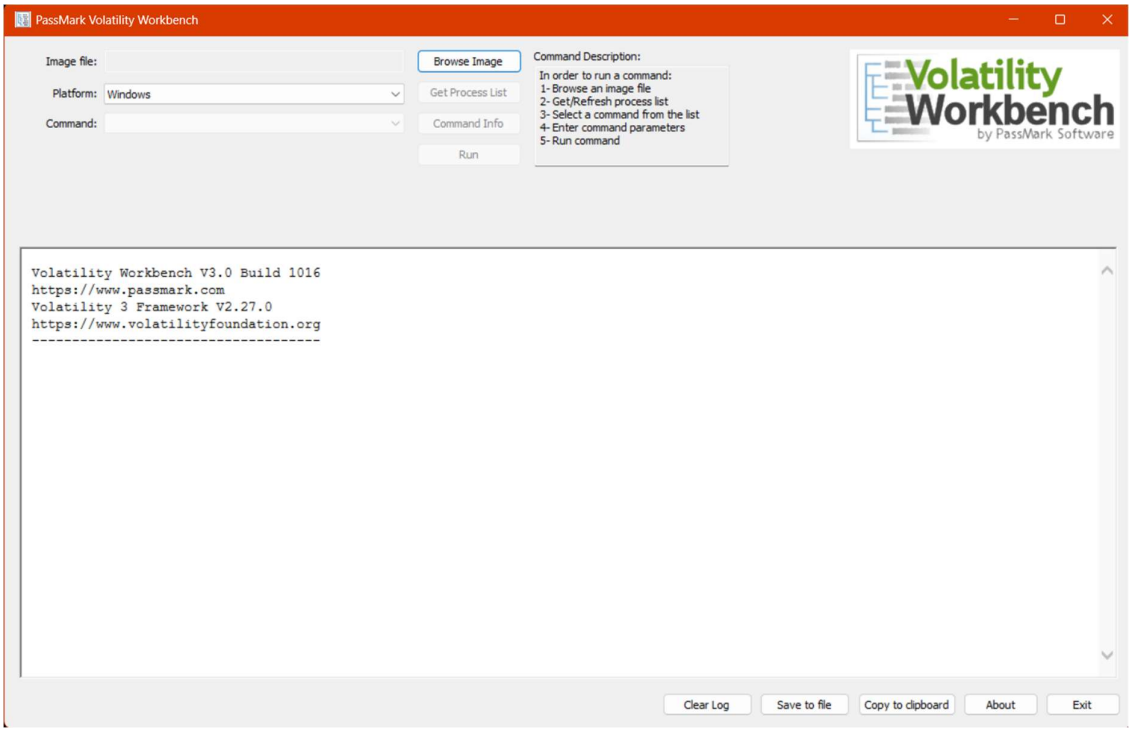


Figure 16: The default interface of PassMark Volatility Workbench

Step 2: Construct and Execute the Volatility Command

To parse the captured memory and list the active processes, load ragul.raw via **Browse Image**, select the **Windows** platform, choose windows.pslist.PsList from the dropdown, and click **Run**.

The console executes the following command in the background:

DOS

"C:\Users\snrag\Downloads\VolatilityWorkbench\vol.exe" -f C:\Users\snrag\Downloads\New folder\ragul.raw" windows.pslist.PsList (Or) Use the workbench

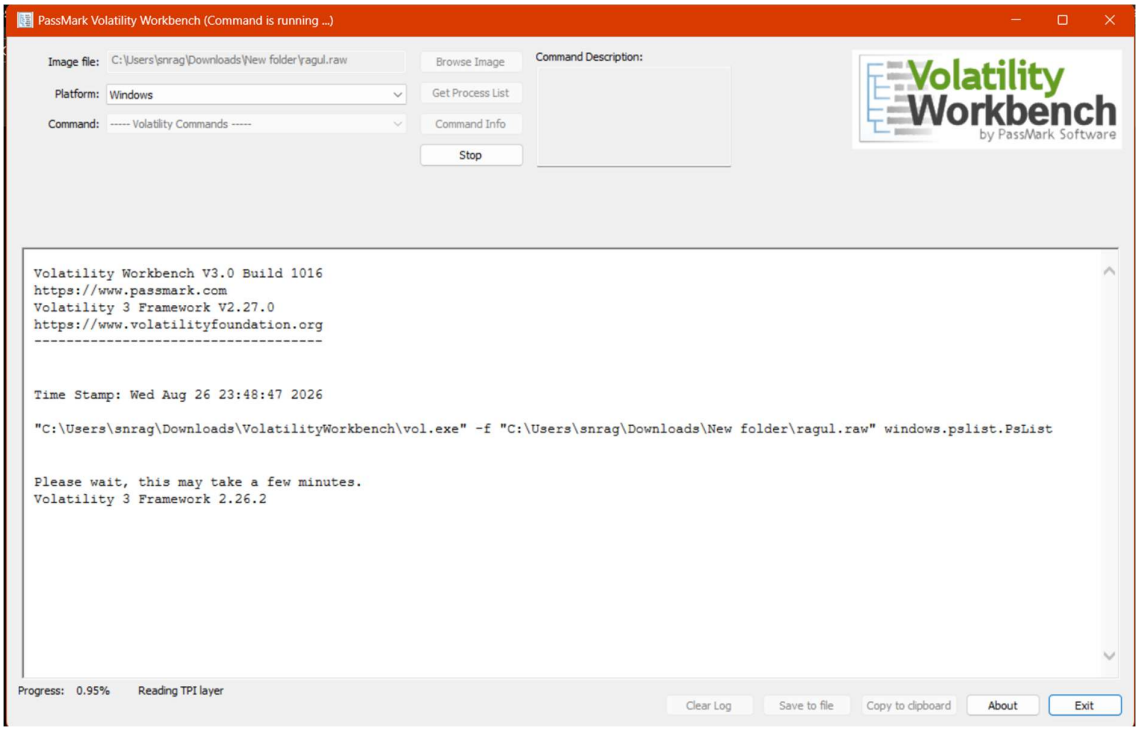


Figure 17:Execution of Volatility for the Inba.raw image

5. OBSERVATIONS & RESULTS

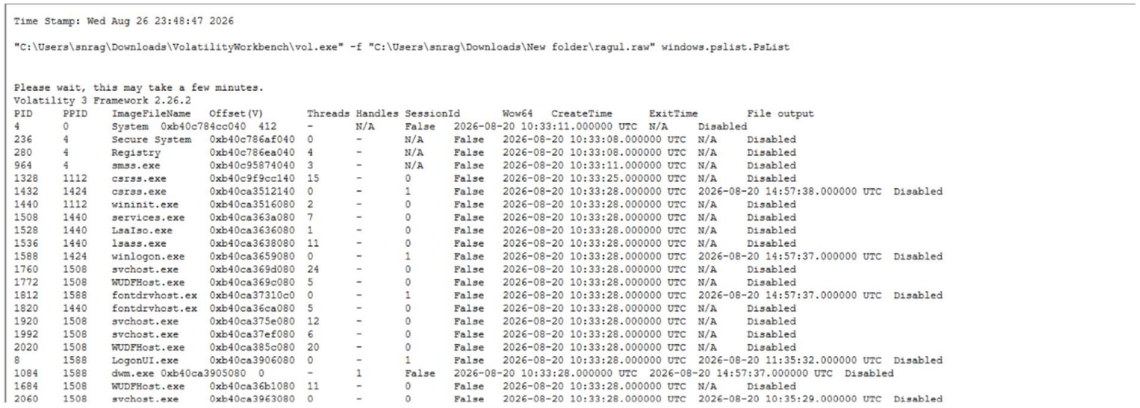
1. FTK Imager Cryptographic Verification

FTK Imager generated a summary report mathematically confirming the non-volatile image data integrity:

- **MD5 Hash:** 3b8d2e0768e0794b8ac45bbacd80e6d2 : verified
- **SHA1 Hash:** f3ae01648923c4a43f097eaal064512317724517a : verified

2. Memory Capture & CLI Parsing

The Inba.raw file was successfully written using the winpmem_mini_x64_rc2.exe command. When analyzed, Volatility parsed the _EPROCESS blocks in kernel memory. The resulting output log accurately detailed the Process IDs (PID), Parent Process IDs (PPID), Image File Names, and timestamps of all active programs:



```
Time Stamp: Wed Aug 26 23:48:47 2026

"C:\Users\snrag\Downloads\VolatilityWorkbench\vol.exe" -f "C:\Users\snrag\Downloads\New folder\ragul.raw" windows.pslist.PsList


Please wait, this may take a few minutes.
Volatility 3 Framework 2.26.2
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output
4       0       System  0xb40c784cc040  412     -       N/A     False   2026-08-20 10:33:11.000000 UTC  N/A     Disabled
236     4       Secure System   0xb40c786af040  0       -       N/A     False   2026-08-20 10:33:08.000000 UTC  N/A     Disabled
280     4       Registry        0xb40c786ea040  4       -       N/A     False   2026-08-20 10:33:08.000000 UTC  N/A     Disabled
964     4       smss.exe        0xb40c95874040  3       -       N/A     False   2026-08-20 10:33:11.000000 UTC  N/A     Disabled
1328    1112    csrss.exe       0xb40c9f9cc140  15      -       0       False   2026-08-20 10:33:25.000000 UTC  N/A     Disabled
1432    1424    csrss.exe       0xb40ca3512140  0       -       1       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 14:57:38.000000 UTC  Disabled
1440    1112    wininit.exe     0xb40ca3516080  2       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1508    1440    services.exe    0xb40ca363a080  7       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1528    1440    LsaIso.exe      0xb40ca3636080  1       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1536    1440    lsass.exe       0xb40ca3638080  11      -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1588    1424    winlogon.exe    0xb40ca3659080  0       -       1       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 14:57:37.000000 UTC  Disabled
1760    1508    svchost.exe     0xb40ca369d080  24      -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1772    1508    WUDFHost.exe    0xb40ca369c080  5       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1812    1588    fontdrvhost.ex  0xb40ca37310c0  0       -       1       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 14:57:37.000000 UTC  Disabled
1820    1440    fontdrvhost.ex  0xb40ca36ca080  5       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1920    1508    svchost.exe     0xb40ca375e080  12      -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
1992    1508    svchost.exe     0xb40ca37ef080  6       -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
2020    1508    WUDFHost.exe    0xb40ca385c080  20      -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
8       1588    LogonUI.exe     0xb40ca3906080  0       -       1       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 11:35:32.000000 UTC  Disabled
1084    1588    dwm.exe 0xb40ca3905080  0       -       1       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 14:57:37.000000 UTC  Disabled
1684    1508    WUDFHost.exe    0xb40ca36b1080  11      -       0       False   2026-08-20 10:33:28.000000 UTC  N/A     Disabled
2060    1508    svchost.exe     0xb40ca3963080  0       -       0       False   2026-08-20 10:33:28.000000 UTC  2026-08-20 10:35:29.000000 UTC  Disabled
```

Figure 18: Volatility Workbench execution log displaying the extracted Process List

Note: The log accurately shows winpmem_mini_x (PID 19448) actively running, reflecting the acquisition tool executing at the time of the capture.

6. CONCLUSION

A complete digital forensics acquisition and analysis workflow was successfully executed via a combination of GUI and CLI tools. The target physical USB drive was securely imaged and verified using FTK Imager. Simultaneously, the system's live RAM was extracted via the Command Prompt using WinPmem. Finally, Volatility 3 command-line syntax was utilized to parse the raw memory file, successfully recovering critical live-state artifacts, including the process list and execution timestamps.